

Image Matching Challenge 2024 - Hexathlon

A Robust SfM Pipeline with LoMa-B Matching and Transparent-Scene Ring Pose Prior

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Abstract

This report presents our final solution for the Kaggle Image Matching Challenge 2024 - Hexathlon. The task is to estimate camera poses for unordered image collections and submit one rotation matrix and one translation vector for each image. Our final system combines a conservative Structure-from-Motion (SfM) pipeline for ordinary scenes with a specialized branch for transparent or reflection-heavy scenes. For normal scenes, we use LoMa-B as the primary local matcher, MAGSAC/RANSAC for geometric verification, and PyCOLMAP incremental reconstruction with shared camera intrinsics. For transparent scenes, where ordinary SfM often fails, we estimate circular image ordering from LoMa and high-resolution ALIKED evidence and assign a ring-pose prior. The best validated Kaggle score is 0.292267, with ablation studies showing that transparent-scene ordering and RANSAC threshold tuning are the two most important improvements.

Code link: `TOD0:addGitHuborGitLabrepositorylink`

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1 Introduction

Image Matching Challenge 2024 - Hexathlon is a camera-pose estimation benchmark built around unordered image collections. Given a set of images from each scene, the goal is to reconstruct camera rotations and translations and write a valid submission file for Kaggle evaluation. Unlike a pure feature-matching task, the final metric depends on pose accuracy, so a system must produce geometrically consistent reconstructions rather than simply maximize the number of matched keypoints.

The main difficulty is scene diversity. Ordinary scenes can often be solved by a strong local matcher followed by robust SfM reconstruction. However, transparent and reflective objects violate the assumptions used by standard multi-view geometry: matches may follow reflections, refractions, or background texture rather than stable object geometry. As a result, a single universal SfM pipeline is fragile.

Our approach addresses this by using two branches. Normal scenes use a LoMa-B and PyCOLMAP pipeline optimized for stable pose estimation. Transparent scenes use a scene-specific ring-pose prior and infer circular image ordering from pairwise matching evidence. This design improves hidden-score performance while keeping the final system reproducible in Kaggle's offline notebook environment.

2 Related Work

2.1 Local Feature Matching

Classical SfM pipelines rely on repeatable keypoints, discriminative descriptors, and robust pairwise matching. Recent learned matchers improve robustness under viewpoint changes, illumination variation, and texture ambiguity. In this project, ALIKED with LightGlue was used as the initial baseline because it is efficient and easy to run offline. LoMa-B was later selected as the primary matcher because it produced stronger correspondence graphs for PyCOLMAP. We also tested GIM, DeDoDe, LoMa-L, RoMa, MAST3R-related variants, and DINO-based pair signals, but these alternatives did not consistently improve the final hidden score.

2.2 Structure-from-Motion

Structure-from-Motion estimates camera poses and sparse 3D points from matched image pairs. A typical SfM system includes geometric verification, camera initialization, triangulation, bundle adjustment, and incremental image registration. We use PyCOLMAP as the final reconstruction backend because it is stable, well-tested, and compatible with the offline Kaggle environment. To reduce unnecessary degrees of freedom, images with identical resolution share camera intrinsics during reconstruction.

2.3 Robust Geometric Verification

Robust verification is essential because incorrect correspondences can damage the pose graph. We use MAGSAC/RANSAC-style fundamental-matrix verification before writing matches into the COLMAP database. The threshold controls a tradeoff between graph connectivity and outlier rejection. Our ablation found that overly strict thresholds remove useful edges, while overly loose thresholds admit harmful outliers. The best validated setting is around 1.075 to 1.08, with the final model using 1.08.

2.4 Transparent and Reflective Scenes

Transparent and reflective scenes are difficult for ordinary image matching and SfM. The apparent texture may come from background objects, reflections, or refractions, so high-confidence local matches may still be geometrically inconsistent. Instead of forcing these scenes through the normal SfM branch, our method estimates circular image ordering and assigns a ring-pose prior. This is motivated by the observation that many transparent-object scenes follow a turntable-like capture pattern.

3 Proposed Approach

3.1 Pipeline Overview

The final system is based on two scene-dependent branches:

1. Normal-scene branch:

Extract and match LoMa-B features, verify correspondences using MAGSAC/RANSAC, build a COLMAP database, share intrinsics for same-resolution images, and reconstruct poses with PyCOLMAP.

2. Transparent-scene branch:

Compute pairwise LoMa and high-resolution ALIKED evidence, combine them as ranking signals, infer circular image ordering, and output ring-prior poses.

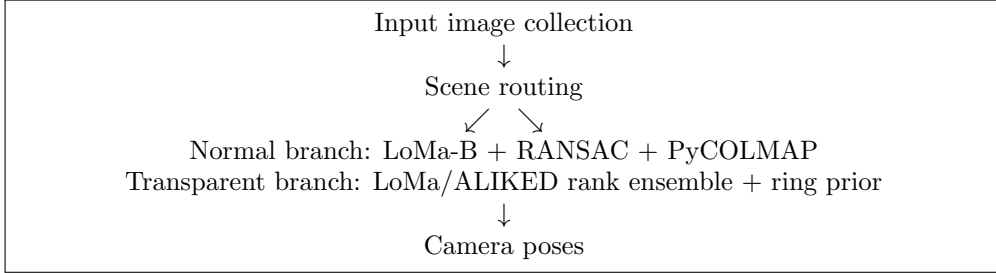


Figure 1: High-level overview of the proposed two-branch pipeline. Replace this placeholder with the final system diagram if needed.

3.2 Normal-Scene SfM Branch

Table 1: Components of the normal-scene reconstruction branch.

Stage	Implementation and purpose
Feature matching	LoMa-B is used as the primary matcher to produce reliable local correspondences.
Geometric verification	MAGSAC/RANSAC with threshold 1.08 filters outliers while preserving enough graph connectivity.
Database construction	Verified matches are written into a COLMAP-compatible database.
Camera model control	Images with identical resolution share camera intrinsics to reduce overfitting.
Reconstruction	PyCOLMAP incremental mapping estimates camera rotations and translations.
Fallback	Scene-level ALIKED + LightGlue is used for weak scenes instead of mixing noisy fallback edges into strong LoMa graphs.
Missing poses	Centroid shift is used only when verified matches are at least 80; otherwise, a conservative pose copy is used.

3.3 Transparent-Scene Branch

The transparent branch is the largest source of improvement. It avoids ordinary SfM and instead assumes a circular camera arrangement. Pairwise matching evidence is converted into ranking signals, then blended with a 20% LoMa and 80% ALIKED weighting. The resulting order is used to assign ring-prior camera poses.

Table 2: Final transparent-scene design choices.

Design choice	Final setting
Scene strategy	Ring-pose prior for transparent and reflection-heavy scenes.
Ordering evidence	LoMa plus high-resolution ALIKED pair evidence.
Rank weights	20% LoMa and 80% ALIKED.
Rejected alternatives	DeDoDe-only ordering, SAM masking, heavy PnP rescue, DINOv2 pair pruning, and aggressive transparent rotation search.

4 Implementation Details

Kaggle reruns final notebooks with internet disabled, so all required wheels, source trees, and model checkpoints must be bundled as a Kaggle Dataset. The notebook installs packages using `--no-index`, copies model weights into local cache paths, and writes the final prediction file to `/kaggle/working/submission.csv`.

Table 3: Important implementation settings.

Parameter	Value
Primary matcher	LoMa-B
SfM backend	PyCOLMAP incremental mapping
RANSAC threshold	1.08
Shared intrinsics	Enabled for images with identical resolution
Transparent branch	Enabled with LoMa/ALIKED rank ensemble and ring-pose prior
Submission output	<code>/kaggle/working/submission.csv</code>

5 Experimental Results

5.1 Evaluation Setup

Each experiment changes a small number of controlled variables and is validated by Kaggle hidden-score submission. The main experiment categories include direct matcher replacement, SfM stability modules, fallback and missing-pose recovery, transparent-scene specialization, and scalar hyperparameter tuning. The final metric is sensitive to camera-pose accuracy, so a version with more registered images is not necessarily better if the estimated geometry is inconsistent.

5.2 Score Progression

Table 4: Major score improvements during development.

Stage	Version	Score	Main reason
ALIKED baseline	ver1	0.201764	Initial ALIKED + LightGlue + COLMAP baseline.
Strong matcher	ver9	0.241186	LoMa-B primary reconstruction.
Shared intrinsics	ver19	0.252244	Reduced camera-parameter instability.
Safe recovery	ver28	0.255489	Gated centroid recovery for missing poses.
Transparent branch	ver71	0.282382	LoMa/ALIKED transparent ring ordering.
RANSAC tuning	ver102	0.288293	RANSAC threshold 1.00.
Final version	ver141	0.292267	RANSAC threshold 1.08.

5.3 Ablation Study

Table 5: Representative ablation results.

Version	Score	Change	Conclusion
ver28	0.255489	Pre-transparent baseline	Strong normal-scene pipeline, but transparent scenes remain weak.
ver71	0.282382	Transparent ring ordering	Largest improvement; scene-specific prior is effective.
ver102	0.288293	RANSAC threshold 1.00	Better connectivity and outlier balance.
ver141	0.292267	RANSAC threshold 1.08	Best validated submission.
ver144	0.289834	Low-weight DeDoDe transparent signal	Competitive but below final version.
ver169	0.290150	RANSAC threshold 1.075	Second-best result; confirms useful threshold region.

5.4 Discussion

The experiments show that the largest gain does not come from simply using a newer or heavier matcher. LoMa-B is a strong primary model, but the breakthrough comes from recognizing that transparent scenes require a different geometric assumption. The ring-pose prior avoids forcing reflection-heavy scenes into an unstable SfM reconstruction. After that, RANSAC threshold tuning provides the strongest secondary improvement by improving the graph connectivity and outlier tradeoff for normal scenes.

Several methods did not improve the final system. GIM, DeDoDe primary matching, LoMa-L, DINOv2 pair pruning, GLOMAP, MAST3R rescue, SAM masking, and heavy PnP recovery either reduced score or introduced runtime and offline-dependency risks. These results suggest that hidden-score robustness is more important than local complexity.

5.5 Error Analysis and Lessons

The ablation log shows that registration count alone was not a reliable indicator of final performance. Some versions produced many matches or registered many images but still scored poorly because the recovered camera geometry was inaccurate. This happened in GIM primary runs, PnP/MASt3R rescue runs, and DeDoDe primary runs. The final pipeline therefore uses conservative missing-pose recovery instead of aggressively inserting low-confidence poses.

The experiments also show that model recency was not a reliable predictor. MAST3R, GLOMAP, DeDoDe, SAM, PixSfM, DINO-based pair selection, and LoMa-L were all plausible upgrades, but most reduced the hidden score in this integrated pipeline. A strong standalone model can still hurt if its feature distribution, pair selection, geometric verification, COLMAP database construction, or offline execution behavior is not calibrated with the rest of the system.

Transparent scenes needed a different objective. They are not simply harder normal scenes: reflections, refractions, and background texture can create visually plausible but physically inconsistent correspondences. The successful branch changed the objective from full SfM reconstruction to plausible circular ordering, which produced the largest score jump.

RANSAC threshold tuning was unusually sensitive. Strict thresholds such as 0.50, 0.90, 0.95, and 0.98 weakened connectivity, while overly loose settings such as 1.085, 1.09, and 1.25 allowed more damaging outliers. The best region was narrow, with 1.075 and 1.08 giving the two best final scores.

5.6 Submission Strategy

The final recommended submission is ver141 because it has the highest validated score and does not depend on fragile optional branches. If multiple submissions are allowed, the useful backups are ver169, ver144, ver136, and ver102. These backups cover nearby RANSAC settings, a low-weight DeDoDe-assisted transparent signal, transparent resize/rotation search, and a simpler strong baseline.

6 Conclusion

We present a robust two-branch camera-pose estimation pipeline for Image Matching Challenge 2024 - Hexathlon. The normal branch uses LoMa-B matching, MAGSAC/RANSAC verification, shared intrinsics, and PyCOLMAP reconstruction. The transparent branch uses LoMa/ALIKED rank evidence and a ring-pose prior to handle scenes where ordinary SfM is unreliable. The final best score is 0.292267. The main findings are that transparent-scene specialization is the dominant improvement and that RANSAC threshold tuning around 1.075 to 1.08 provides the best secondary gain.

Team Member Contribution

Table 6: Team member contribution table. Update names and percentages before final submission.

Student	Survey	Design	Impl.	Report	Slides
112550200 Zheng Wu Qian	25%	25%	25%	25%	25%
112550103 Xiang-Yu You	25%	25%	25%	25%	25%
112550201 Team Member Name	25%	25%	25%	25%	25%
112550127 Pin-Kuan Chiang	25%	25%	25%	25%	25%

References

- [1] Kaggle. Image Matching Challenge 2024 - Hexathlon.
- [2] J. L. Schonberger and J.-M. Frahm. Structure-from-Motion Revisited. In *CVPR*, 2016.
- [3] D. Barath, J. Matas, and J. Nuskova. MAGSAC: Marginalizing Sample Consensus. In *CVPR*, 2019.
- [4] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys. LightGlue: Local Feature Matching at Light Speed. In *ICCV*, 2023.
- [5] X. Zhao, X. Wu, W. Chen, P. C. Y. Chen, Q. Xu, and Z. Li. ALIKED: A Lighter Keypoint and Descriptor Extraction Network via Deformable Transformation. *IEEE Transactions on Instrumentation and Measurement*, 2023.
- [6] D. Nordstrom, J. Edstedt, G. Bokman, G. Astermark, J. Heyden, A. Larsson, V. Wadenback, M. Felsberg, and F. Kahl. LoMa: Local Feature Matching Revisited. arXiv preprint, 2026.
- [7] J. Edstedt, G. Bokman, V. Wadenback, and M. Felsberg. DeDoDe: Detect, Don't Describe – Describe, Don't Detect for Local Feature Matching. In *3DV*, 2024.
- [8] X. Shen, Z. Cai, W. Yin, M. Muller, Z. Li, K. Wang, X. Chen, and C. Wang. GIM: Learning Generalizable Image Matcher From Internet Videos. In *ICLR*, 2024.